RAMDEOBABA UNIVERSITY, NAGPUR

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B.Tech. CSE (Artificial Intelligence and Data Science) (AIDS)



SEMESTER V

SI LAB

PROJECT REPORT

<<INSERT PROJECT TITLE HERE>>

Submitted by

Student Name	Roll No.	Enrollment No.

Academic Year: 2026–27

<< PROJECT TITLE >>

SI Lab Project Report

1. PROJECT OBJECTIVE

State the main purpose of the project, the problem it addresses, the expected output, and the intended use.

Suggested objectives for a classification project:

- Load and explore the selected dataset.
- Preprocess the data and prepare it for model building.
- Extract or select relevant features, where applicable.
- Train multiple classification models.
- Evaluate and compare model performance.
- Select the best model using suitable evaluation metrics.
- Develop a GUI to accept input and display predictions.

2. INTRODUCTION

Introduce the project domain, background, motivation, problem being addressed, and expected usefulness.

2.1 Problem Statement

Clearly define the problem the project aims to solve.

2.2 Scope of the Project

Describe the functions included in the project and its limitations.

3. PROJECT METHODOLOGY

Explain the stages followed in the project.

3.1 Dataset Loading and Exploration

- Dataset name and source/link.
- Number of rows and columns, features, target variable and data types.
- First five rows, dataset information, missing values and duplicate records.
- Class distribution and relevant exploratory visualizations.

3.2 Data Preprocessing

- Handling missing values and duplicates, if required.
- Encoding categorical variables, if required.
- Feature scaling/normalization, if required.
- Separating input features and target variable.
- Train-test split and other preparation steps.

3.3 Feature Extraction / Selection

Describe the features used and any feature extraction/selection technique applied. If no separate feature extraction is needed, explain how the available features are used.

3.4 Training Multiple Classification Models

List the classification models implemented and briefly explain their purpose.

Sr. No.	Model Name	Purpose / Notes
1	<< Model 1 >>	<< Brief explanation >>
2	<< Model 2 >>	<< Brief explanation >>
3	<< Model 3 >>	<< Brief explanation >>
4	<< Model 4 >>	<< Brief explanation >>

3.5 Performance Evaluation and Model Comparison

Use suitable classification metrics such as accuracy, precision, recall, F1-score and confusion matrix, as appropriate.

Model	Accuracy	Precision	Recall	F1-Score
<< Model 1 >>	<< actual >>	<< actual >>	<< actual >>	<< actual >>
<< Model 2 >>	<< actual >>	<< actual >>	<< actual >>	<< actual >>
<< Model 3 >>	<< actual >>	<< actual >>	<< actual >>	<< actual >>

Replace all placeholders with results generated by your own program. Do not copy values from another project.

3.6 Best Model Selection

Identify the selected model and justify the choice using the evaluation results and project requirements.

4. PROJECT IMPLEMENTATION

4.1 Tools and Technologies

Tool / Technology	Version	Purpose
Python	<< Version >>	Programming language
Google Colab	<< >>	Development environment
Pandas / NumPy	<< >>	Data handling
Scikit-learn	<< >>	Preprocessing and models
Matplotlib / Seaborn	<< >>	Visualization
GUI framework	<< Name >>	User interface

4.2 Module-wise Implementation

- Dataset loading and exploration module.
- Data preprocessing module.
- Feature extraction/selection module, if applicable.
- Model training and evaluation module.
- Best model saving/loading module.
- GUI input and prediction module.

4.3 GUI Implementation

Describe the GUI layout, input fields, prediction process, and how the selected model generates predictions. Include screenshots of the interface and prediction output.

5. COMPLETE CODE

Include the complete runnable source code in this section or provide it through the Colab notebook and GitHub repository. Organize code into logical sections and add comments where helpful. Code must correspond to the report's results and screenshots.

6. RESULTS AND DISCUSSION

Present actual outputs from the implemented project. Give each screenshot a figure number, caption and short explanation.

6.1 Required Output Screenshots

- Data preprocessing output.
- Model comparison table.
- Accuracy/performance results.
- Confusion matrix.
- Best model selection.
- GUI interface.
- GUI prediction results.

6.2 Result Analysis

Explain key observations, differences between models, strengths, errors, limitations, and whether the objectives were achieved.

7. CONCLUSION

Summarize the project, models/methods implemented, main results and final selected model. Mention key limitations and possible improvements.

8. GITHUB LINK

GitHub Repository URL: << Paste the repository URL here >>

The repository should contain Python code files, dataset or dataset link, trained model file, GUI application code and Colab notebook.

9. REFERENCES

List the dataset source, official documentation, libraries/frameworks, websites, books and other sources actually used. Use a consistent format.

[1] << Dataset name and URL >>

[2] << Official library/documentation URL >>

[3] << Other source used >>

FINAL SUBMISSION CHECKLIST

- Final project report in PDF format.
- Google Colab notebook (.ipynb).
- Colab notebook exported as PDF.
- GitHub repository containing Python code, dataset or dataset link, trained model file, GUI code and Colab notebook.

Formatting: The first page should contain college/university name, department, logo, project title and student details. From page 2 onwards, use project title in Times New Roman 16 pt, headings/subheadings in Times New Roman 14 pt, and normal text in Times New Roman 12 pt, justified.

Replace all << placeholders >> with actual project information. Verify that all screenshots and numerical results are from your own implementation.